

BURUKAN, Russian Federation

WMO: 313480

Lat: **53.050N**Lon: **136.033E**Elev: **153**StdP: **99.50**Time Zone: **10.00 (E10)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -43.5	(c) -41.3	(d) -45.5	(e) 0.0	(f) -42.8	(g) -43.5	(h) 0.1	(i) -40.8	(j) 5.9	(k) -10.9	(l) 5.0	(m) -11.8	(n) 0.1	(o) 350	(p) 0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	28.8	20.0	26.8	19.0	24.7	17.9	21.6	26.8	20.3	25.0	19.0	23.2	1.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.8	24.2	18.5	13.6	22.8	17.2	12.5	21.5	63.9	26.3	59.2	25.0	54.7	23.3	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.4	(b) 3.8	(c) 3.2	(e) -45.2	(f) 32.6	(g) 2.3	(h) 1.7	(i) -46.8	(j) 33.8	(k) -48.2	(l) 34.8	(m) -49.4	(n) 35.8	(o) -51.1	(p) 37.1		
(4) 4.4	(5) 3.8	(6) 3.2	(7) -45.1	(8) 23.7	(9) 2.0	(10) 1.3	(11) -46.5	(12) 24.6	(13) -47.6	(14) 25.4	(15) -48.8	(16) 26.1	(17) -50.2	(18) 27.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.7	-26.9	-22.8	-13.0	-1.3	6.5	12.9	16.3	15.7	9.2	-0.5	-15.4	-26.3		
	DBStd	16.58	6.66	5.66	6.00	4.73	3.99	3.96	3.18	3.00	3.59	4.64	7.85	6.57		
	HDD10.0	5524	1143	920	713	339	123	16	1	1	56	326	762	1124		
	HDD18.3	8077	1402	1153	971	589	368	169	81	92	274	584	1012	1382		
	CDD10.0	522	0	0	0	1	14	102	195	178	32	0	0	0		
	CDD18.3	32	0	0	0	0	0	6	16	10	0	0	0	0		
	CDH23.3	720	0	0	0	1	28	212	299	169	11	0	0	0		
	CDH26.7	172	0	0	0	0	6	57	81	28	1	0	0	0		
(14) Wind	WSAvg	1.0	0.7	0.7	1.1	1.3	1.4	1.2	1.0	0.9	0.9	0.9	0.8	0.6		
(15) Precipitation	PrecAvg	642	13	11	18	49	60	75	108	115	94	55	26	20		
	PrecMax	932	51	35	46	109	126	180	222	262	169	173	102	88		
	PrecMin	387	0	0	1	1	10	20	20	25	20	8	0	0		
	PrecStd	127	14	10	12	25	29	35	58	60	39	33	23	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.6	-4.0	6.7	19.3	27.6	31.1	32.1	29.5	24.7	15.7	2.5	-2.8		
		MCWB	-5.0	-6.2	1.6	9.6	15.2	20.8	21.8	21.4	17.8	9.5	-0.2	-3.0		
	2%	DB	-8.0	-7.4	3.0	13.6	23.0	28.3	29.2	27.1	21.6	11.6	0.0	-7.7		
		MCWB	-8.4	-9.1	-1.1	6.5	12.7	18.6	20.7	20.0	15.5	6.5	-1.3	-8.2		
	5%	DB	-11.6	-9.8	0.2	10.4	20.1	25.8	26.9	25.2	19.3	9.1	-1.5	-11.5		
		MCWB	-12.0	-10.9	-2.7	4.7	11.9	17.4	19.6	19.2	14.3	5.3	-2.7	-12.0		
	10%	DB	-15.0	-12.1	-1.9	7.5	16.7	23.3	24.5	23.2	17.0	7.1	-3.2	-14.8		
		MCWB	-15.5	-13.0	-4.2	2.7	10.1	16.1	18.6	17.9	12.7	4.1	-4.2	-15.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.9	-6.0	1.9	10.4	16.5	22.6	23.5	22.8	18.9	10.2	0.6	-2.9		
		MCDB	-4.5	-4.1	6.4	17.7	24.5	29.0	29.0	26.8	23.5	14.9	1.6	-2.8		
	2%	WB	-8.4	-9.0	-0.8	7.3	13.9	20.1	21.7	21.3	16.6	7.7	-1.1	-8.5		
		MCDB	-8.1	-7.5	2.3	13.2	21.1	26.3	27.3	25.8	20.1	10.1	-0.1	-8.0		
	5%	WB	-12.0	-10.9	-2.4	4.9	12.0	18.3	20.5	20.0	15.0	6.0	-2.5	-12.2		
		MCDB	-11.6	-9.9	0.0	10.0	19.2	24.3	25.5	23.9	18.1	8.7	-1.6	-11.7		
	10%	WB	-15.4	-13.0	-4.1	2.9	10.3	16.6	19.1	18.7	13.5	4.4	-4.2	-15.4		
		MCDB	-14.9	-12.1	-2.0	7.0	16.3	22.1	23.5	22.0	16.3	6.8	-3.1	-14.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	19.0	18.4	13.7	13.6	14.6	12.0	12.3	14.5	12.5	14.0	14.2		
		MCDBR	11.5	18.7	20.1	19.4	22.7	22.3	19.4	17.9	19.5	16.0	11.8	11.0		
		MCWBR	11.0	17.0	16.3	12.7	13.5	13.1	11.3	11.2	13.0	11.0	9.9	10.4		
		MCDBR	10.8	18.2	18.6	18.5	21.1	19.4	16.4	15.2	16.6	14.4	10.5	10.5		
	5% WB	MCWBR	10.4	16.5	14.9	12.4	13.4	12.2	10.3	10.3	11.7	10.4	9.0	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.255	0.287	0.352	0.461	0.532	0.493	0.498	0.425	0.368	0.360	0.295	0.255		
	taud		2.135	2.088	1.937	1.781	1.807	2.034	2.063	2.287	2.385	2.193	2.109	2.082		
	Ebn at Noon		719	804	808	758	726	767	753	796	806	704	640	627		
	Edh at Noon		68	99	144	194	199	161	154	117	93	91	70	59		
(44) All-Sky Solar Radiation	RadAvg		1.28	2.44	4.01	5.11	4.92	5.40	4.84	4.11	3.21	2.00	1.28	0.95		
	RadStd		0.06	0.10	0.22	0.32	0.57	0.53	0.53	0.45	0.29	0.17	0.11	0.05		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
											(47)

Nomenclature: See separate page